Predicting Answer Correctness from Reasoning Traces

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Abstract

Large Language Models (LLMs) increasingly rely on Chain-of-Thought (CoT) reasoning to solve complex problems. However, the correctness of the final answer is often unknown until the generation is complete. We propose to predict whether a small reasoning LLM will answer correctly *before* it finishes inference by training lightweight classifiers on hidden states collected from reasoning tokens. We systematically study *where* (in which layers) and *when* (at which tokens) the "correctness" signal emerges in small (1.5B) reasoning models, specifically comparing a reasoning-native model (DeepSeek-R1-Distill-Qwen-1.5B) and an instruction-tuned model (Qwen2.5-Math-1.5B). We evaluate our probes on GSM8K and MATH datasets, judging performance by sequence-level accuracy and AUROC. Furthermore, we investigate the transferability of these probes across datasets and model families. Our work aims to provide a computationally efficient method for early error detection and reliability enhancement in resource-constrained settings. For reproducibility, we release all code for our experiments at `https://github.com/cck0517/model_probe`

1 Introduction

The deployment of Large Language Models (LLMs) in safety-critical or resource-constrained environments necessitates a reliable mechanism to gauge the trustworthiness of their outputs. While Chain-of-Thought (CoT) prompting (Wei et al., 2022) has significantly improved reasoning capabilities, it also introduces a longer inference latency. In many practical applications, it is valuable to know *before the final answer is generated* whether the reasoning process is proceeding towards a correct conclusion. If a reliable correctness signal can be detected early in the model's hidden states, we can enable several powerful capabilities: (i) **early-exit**, stopping generation to save compute when the model is confident of failure or success; and (ii) **selective verification**, flagging low-confidence traces for human review or external tool use.

Problem Formulation Formally, given an input prompt x and a reasoning model $\mathcal{M}$, the model generates a sequence of reasoning tokens (the "thought" or "rationale") $t_{1:T}$ followed by a final answer y. Our goal is to learn a lightweight classifier $g(z_{\ell,f}) \to \{0, 1\}$ that predicts whether the final answer y will be correct, based on a pooled feature vector $z_{\ell,f} \in \mathbb{R}^d$ extracted from layer ℓ under feature definition f (e.g., reasoning-mean, reasoning-last). Rather than probing individual token activations $h_{\ell,k}$, we use aggregated representations of the reasoning span, which are more practical for large-scale extraction and closer to deployment constraints.

Scope and Contributions While prior work has explored self-evaluation (Kadavath et al., 2022) and probing for truthfulness (Zhang et al., 2025), our work focuses specifically on the dynamics of correctness signals in small, distilled reasoning models (1.5B parameters). These models are increasingly important for on-device and edge applications where efficiency is paramount. We make the following contributions:

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1. **Systematic Signal Analysis**: We map the emergence of correctness signals across layers and token positions, identifying the "critical path" where the model internally resolves the answer.

2. **Model Comparison**: We contrast the internal representations of a "reasoning-native" model (DeepSeek-R1), which was trained with explicit reasoning objectives, against a standard instruction-tuned model (Qwen2.5) prompted to reason. This sheds light on whether explicit reasoning training leads to more structured internal states.

3. **Transfer Learning Evaluation**: We test the robustness of our probes by evaluating their performance across distribution shifts (e.g., training on GSM8K and testing on MATH) and across model architectures.

Taken together, these studies let us ask a focused question about the reliability signals encoded by small reasoning models. In this work, we test the following hypothesis:

Hypothesis. *Small (1.5B) reasoning models encode a linearly decodable correctness signal in their hidden states that emerges along the chain-of-thought trajectory and is strong enough to support early error detection across datasets and model families.*

2 Related Work

The study of internal representations in large language models (LLMs) has exploded in scope, spanning prompting strategies, self-evaluation, interpretability, and uncertainty quantification. Our work sits at the intersection of these lines of research: we probe hidden states of *small* (1.5B) reasoning models to predict answer correctness from the reasoning tokens. In this section, we review: (i) CoT reasoning and external verification, (ii) self-evaluation and calibration, (iii) probing and decoding of internal states, (iv) internal-state approaches to hallucination and uncertainty, (v) correctness prediction from activations, and (vi) reasoning distillation and small models.

2.1 Chain-of-Thought, Self-Consistency, and External Verification

Chain-of-Thought prompting (Wei et al., 2022) showed that eliciting step-by-step rationales dramatically improves performance on math and logical reasoning benchmarks such as GSM8K and MATH (Cobbe et al., 2021; Hendrycks et al., 2021). By viewing the model's generation as an explicit sequence of intermediate computations rather than a single-step mapping from question to answer, CoT exposes a rich trajectory of hidden activations—precisely the object we study in this work. Self-Consistency (Wang et al., 2022) further improves accuracy by sampling multiple diverse reasoning paths and selecting the answer by majority vote. This suggests that correct reasoning trajectories occupy a larger or more stable basin in model space than incorrect ones, motivating the idea that there may be a detectable "correctness" signature in internal states.

A parallel line of work attaches explicit verifiers to CoT solutions. Cobbe et al. (2021) introduce GSM8K and train separate neural verifiers that score candidate solutions; at test time, the final answer is chosen by verification rather than raw generation probability. Subsequent work scales up this idea, e.g., Math-Shepherd, which trains a process reward model to score each intermediate step of math solutions and uses it as a reward model for reinforcement learning (Wang et al., 2024). Achieving $> 97\%$ accuracy on GSM8K via deep problem understanding (Zhong et al., 2024) also relies on external reasoning pipelines that encourage models to explicitly parse and structure problem information before solving it. These approaches, however, typically treat the base LM as a black box and rely on additional sampling or external verifiers, which can be expensive.

Process supervision pushes verification inside the reasoning trajectory. Lightman et al. (2023) introduce process reward models trained with human labels over intermediate reasoning steps, showing that process supervision is more sample-efficient than outcome supervision on MATH. Related work on self-verification (Weng et al., 2023) and collaborative verification schemes also leverage the idea that models can critique or verify their own CoT. Our work is complementary: instead of training separate verifiers or reward models, we ask whether a simple linear classifier over internal activations can predict whether a given CoT *will* end in a correct answer, and whether this signal emerges early enough to enable early exit.

2.2 Self-Evaluation, Confidence, and Calibration

A growing body of work studies "self-knowledge" in LLMs: whether models can assess the correctness of their own outputs. Prior work (Kadavath et al., 2022) shows that models can often provide reasonably calibrated natural-language estimates of the probability that their answers are true (*P(True)*), and that this signal can outperform raw log-probabilities as a confidence estimate. Follow-up work (Weng et al., 2023) demonstrates that large models with CoT can act as better reasoners when explicitly prompted to self-verify their conclusions, by performing a backward check of the CoT steps and assigning interpretable validation scores. More recent work explores training objectives that directly incentivize self-verification during fine-tuning or reinforcement learning, showing improvements on math and logical reasoning benchmarks.

These self-evaluation methods are typically *verbalized*: the model generates extra text that expresses confidence, critiques, or alternative solutions. This requires additional decoding budget and introduces its own failure modes (e.g., the model confidently misjudging its own errors). In contrast, our probes operate on hidden states extracted during the normal CoT forward pass, without requiring extra generations or auxiliary prompts. Conceptually, we treat self-knowledge as a latent feature in the activation space and attempt to decode it *directly*, rather than having the model explain itself in natural language.

Calibration and uncertainty estimation approaches based on output distributions are another closely related thread. Semantic-entropy estimators (Farquhar et al., 2024) measure uncertainty over diverse generations, showing strong performance in detecting confabulations in factual QA. Other work uses Bayesian decision theory or sequential tests to adaptively decide when enough evidence has been gathered to trust an output. These methods operate at the logit or sequence level; our work instead leverages intermediate hidden states, which can potentially reveal confidence signals earlier in the computation and in a format amenable to linear probing.

2.3 Hidden-State Probing and Decoding

Linear probing is a core tool in representation analysis. The classic "logit lens" and its refinement, the Tuned Lens (Belrose et al., 2023), train affine probes at each layer of a frozen transformer to decode the current hidden state into a distribution over the vocabulary. These works show that transformers generally refine predictions layer-by-layer, and that the latent predictions extracted by a tuned lens track the model's eventual output distribution more faithfully than the raw logits from early layers. This suggests that important high-level concepts—including correctness—may be encoded linearly in the residual stream.

Beyond next-token prediction, a wide literature applies linear probes to decode syntax, world knowledge, and other abstract properties from activations. Recent sparse autoencoder (SAE) work from Anthropic (Anthropic, 2024) decomposes activations into thousands of monosemantic features, some of which correspond to surprisingly human-interpretable concepts (e.g., specific entities, topics, or stylistic attributes). Follow-up work uses SAEs as a basis for concept steering and monitoring, reinforcing the idea that many global properties of a model's behavior are linearly encoded in its internal representations.

Our use of logistic regression probes is in line with this literature but targets a very specific scalar property: "will this CoT produce a correct final answer?". Compared to general-purpose decoding (e.g., Tuned Lens) or unsupervised SAE feature discovery, our probes are directly supervised with correctness labels and evaluated for prediction and calibration quality. We also analyze *where* (across layers) and *when* (across reasoning tokens) this correctness signal becomes most linearly separable in small models, which is less explored than in the large-model regime studied by most interpretability work.

2.4 Internal Uncertainty and Hallucination Detection

Internal-state–based hallucination detection is a close conceptual neighbor to our approach, as both aim to extract reliability signals *before* (or beyond) the final decoded answer. INSIDE (Chen et al., 2024a) proposes **EigenScore**, an uncertainty metric computed from the *dispersion of internal sentence embeddings* across multiple sampled responses. Concretely, given an input x, one samples K candidate responses $\{y^{(i)}\}_{i=1}^{K}$ and extracts a sentence embedding $z_i \in \mathbb{R}^d$ for each response

from an internal layer (e.g., a middle-layer last-token embedding). Stacking these embeddings into $Z = [z_1, \ldots, z_K] \in \mathbb{R}^{d \times K}$, INSIDE forms a centered covariance matrix (up to a centering transform) $\Sigma \in \mathbb{R}^{K \times K}$ that captures how similar the *internal* representations of different responses are. EigenScore is then defined as the (regularized) log-determinant of this covariance, equivalently the average log-eigenvalue:

$$\text{EigenScore}(x) \;=\; \frac{1}{K} \log\det(\Sigma + \alpha I) \;=\; \frac{1}{K} \sum_{i=1}^{K} \log \lambda_i,$$

where $\{\lambda_i\}$ are eigenvalues of the regularized covariance and α ensures full rank. Intuitively, when the model is confident and produces semantically consistent answers, the embeddings are highly correlated and the spectrum is concentrated (small EigenScore); when it is uncertain or hallucinates, embeddings diversify and the spectrum spreads (large EigenScore). INSIDE further explores test-time feature clipping to dampen extreme activations and reduce overconfident hallucinations.

Subsequent work reinforces the general message that internal signals can be strong predictors of reliability, including approaches that combine hidden activations with additional cues (e.g., attention or output statistics) for hallucination detection (Chen et al., 2024a; Sriramanan et al., 2024). Complementary output-level methods, such as semantic entropy (Farquhar et al., 2024), estimate uncertainty by sampling multiple completions and measuring semantic diversity after decoding.

Our setting differs in two key ways. First, we target *symbolic/math reasoning* (GSM8K, MATH), where errors often stem from incorrect intermediate computations rather than factual misalignment with an external knowledge base. Second, we emphasize *single-trace* monitoring in small (1.5B) models: instead of requiring K sampled responses to form a covariance spectrum, we ask whether a lightweight probe over hidden states from one CoT trace can predict eventual correctness early enough to support practical early-exit or selective verification.

2.5 Predicting Correctness in Reasoning Models

Most directly related to our work are methods that probe internal states to predict answer correctness. Zhang et al. (2025) study large reasoning-specialized models and train probes on hidden states at selected intermediate reasoning positions to predict whether the final answer will be correct. They report strong separability (e.g., AUROC) and show that the signal can support early-exit policies with minimal loss in final accuracy, with correctness evidence often peaking in mid-to-late layers.

A complementary line of work (Cencerrado et al., 2025) asks how early correctness can be predicted *before* reasoning begins: they extract activations immediately after the question is read (question-only representations) and train linear probes to predict whether the forthcoming answer will be correct. These probes can generalize well on knowledge or trivia-style benchmarks and sometimes outperform black-box confidence estimates. However, their performance degrades on tasks where difficulty is dominated by multi-step reasoning rather than question surface features.

Our work differs from these correctness-prediction papers along four concrete axes.

(i) Model regime and deployment constraints. Prior studies primarily focus on larger (7B–70B) models and often assume access to token-level activations at many positions. We instead target small 1.5B models that are relevant for on-device/edge settings, and we design our feature extraction to be storage- and compute-aware.

(ii) What is being predicted and when. Question-only probes predict difficulty from the prompt alone, while intermediate-position probes typically choose a small number of reasoning checkpoints. In contrast, we study correctness signals *along the reasoning trajectory* by comparing features that summarize different portions of the CoT (e.g., aggregates over the full reasoning span vs. prefix-only aggregates). This lets us test whether correctness evidence strengthens over the course of reasoning and whether "earlier" summaries suffice for monitoring.

(iii) Representation choice: pooled reasoning features vs. token-level probes. While token-level probing can provide fine-grained temporal localization, it can be expensive to store and process at scale. We therefore evaluate lightweight probes over fixed-size pooled representations (e.g., reasoning-mean, reasoning-last, and prefix-based summaries), which are closer to a realistic monitoring interface in deployed systems. This design also enables systematic sweeps over layers and feature definitions without storing full activation tensors for every token.

(iv) Generalization questions beyond in-distribution monitoring. Existing work emphasizes in-distribution early-exit and separability; we additionally stress transfer: (a) dataset shift (e.g., GSM8K $\rightarrow$ MATH), and (b) model-family differences (reasoning-native vs. prompted-CoT). This directly tests whether correctness directions behave like idiosyncratic heuristics tied to a training distribution/model, or whether they reflect more universal structure in small-model representations.

Beyond activation-based methods, other work studies self-evaluation via verbalized confidence or self-verification (Weng et al., 2023), often requiring extra decoding passes. Our probes instead operate on hidden states from the normal forward computation of a single CoT trace, making them compatible with low-latency monitoring and potentially earlier detection.

2.6 Distilled Reasoning Models and the Small-Model Regime

Recent work shows that multi-step reasoning behaviors can be compressed into smaller models via distillation or process-supervised training (e.g., distilling step-by-step rationales from a teacher, or using step-level reward signals) (Hsieh et al., 2023; Chen et al., 2024b; Wang et al., 2024). Industrial releases such as DeepSeek-R1 distilled variants (DeepSeek-AI, 2025) and compact instruction-tuned math models such as Qwen2.5-Math-1.5B (Qwen Team, 2024) reflect a broader trend toward efficient reasoning in the sub-2B regime.

We do *not* study distillation algorithms directly; rather, we use distilled and non-distilled 1.5B models as a controlled testbed to ask a reliability question that becomes especially salient at small scale: whether correctness signals remain linearly decodable from the hidden states of a single CoT trace, and how these signals differ between reasoning-native training (with explicit reasoning control tokens) and prompted-CoT behavior.

3 Methodology

3.1 Models

We focus on the 1.5B parameter scale to explore the capabilities of lightweight models suitable for edge deployment.

- **DeepSeek-R1-Distill-Qwen-1.5B**: A model distilled from DeepSeek-R1, trained with explicit reasoning data and utilizing specific control tokens (e.g., `<think>`) to demarcate the reasoning process.
- **Qwen2.5-Math-1.5B**: A math-specialized variant of Qwen2.5-1.5B, instruction-tuned on step-by-step mathematical corpora (e.g., GSM8K- and MATH-style problems), but without explicit reasoning control tokens, so we elicit chain-of-thought behavior purely via prompting.

3.2 Datasets and Data Collection

We utilize two standard mathematical reasoning benchmarks:

- **GSM8K**: A dataset of grade-school math word problems.
- **MATH**: A more challenging dataset containing high-school and competition-level math problems.

We adapt the `lm-eval-harness` framework for data collection (EleutherAI, 2020). For GSM8K, we use the official configuration (`task = gsm8k`, `num_fewshot = 5`) with the template `"Question: {{question}}\nAnswer:"` and `output_type = generate_until`. This yields a 5-shot prompt where each in-context example contains a worked-out solution followed by a final line of the form `"#### a"`, so the evaluated question is presented in a chain-of-thought style. The harness performs greedy decoding (`do_sample = false`, `temperature = 0.0`) with an `until` stopping condition (`"Question:"`, `"</s>"`, `"<|im_end|>"`), extracts a numeric answer using the standard regex filters, and computes the `exact_match` metric after light normalization (removing commas, dollar signs, the `"#### "` prefix, and trailing periods).

To attach activations to these labels, we first run `lm-eval-harness` and save its per-example JSONL logs, which contain the instantiated prompt, generation arguments, filtered response, and `exact_match` score. Our extraction script replays the same prompt and generation parameters in `transformers` (reproducing the `until` behavior with a custom stopping criterion), re-computes the answer with the same GSM8K/MATH post-processing, and discards any example where the replayed answer or correctness disagrees with the original `lm-eval` result. For the remaining examples, we concatenate the prompt and reproduced response, run a separate forward pass with `output_hidden_states = True` and `use_cache = False`, and identify the reasoning span as the tokens between the end of the prompt and the answer delimiter ("`####`" for GSM8K). We refer to all generated tokens between the end of the prompt and this answer delimiter as *reasoning tokens*.

To keep storage manageable, we do not store per-token activations for all layers. Instead, for each model we select a small set of depth-proportional layers (e.g., $\lfloor 0.25L \rfloor$, $\lfloor 0.5L \rfloor$, $\lfloor 0.75L \rfloor$ where L is the total number of layers) and compute reasoning-focused pooled features per selected layer: (i) the last reasoning token, (ii) the mean over all reasoning tokens, and (iii) prefix features given by the last and mean activations over the first 25%, 50%, and 75% of the reasoning span. We also record simple sequence-level reference features (last token, mean-pooled, max-pooled over the full prompt+response sequence). These fixed-size pooled vectors are the inputs to our linear probes and make it feasible to analyze thousands of examples without storing the full activation tensor for every token and layer. **We do not claim that these layer positions and pooling schemes are optimal: they were chosen heuristically under strict compute and storage constraints, which prevented a systematic hyperparameter sweep. Instead, we prioritized broad coverage across depths and pooling types, and we view a more exhaustive search over layer and pooling configurations as an important direction for future work.**

Table 1: Base GSM8K performance of the two 1.5B models under the official `lm-eval-harness` 5-shot configuration (`task=gsm8k`, `num_fewshot=5`, greedy decoding). Metrics are exact match (EM) in %.

Model	Params	# Test Qs	EM (strict)	EM (flexible)
DeepSeek-R1-Distill-Qwen-1.5B	1.78B	1319	67.2	67.7
Qwen2.5-Math-1.5B	1.54B	1319	73.5	73.5

3.3 Probing Architecture

We train one probe per (layer, feature) pair, where each feature is a pooled summary of the hidden states over the reasoning span—e.g., the last reasoning token, the mean over all reasoning tokens, or prefix-based aggregates. Thus, the probe input is a single fixed-size vector $z_{\ell,f} \in \mathbb{R}^d$ derived from layer ℓ under feature definition f. In principle, one could train a probe on every token or on full sequence-level activations, but this would introduce significant storage and compute overhead. For practical deployment and for tractable large-scale extraction, we instead use aggregated features.

Each probe is a logistic regression classifier of the form

$$P(\text{Correct} \mid z_{\ell,f}) = \sigma(w_{\ell,f}^\top z_{\ell,f} + b_{\ell,f}),$$

where σ is the sigmoid function. This simple linear form ensures that recovered signal reflects linearly accessible information in the representation, rather than probe expressiveness.

3.4 Evaluation Metrics

- **Accuracy**: The percentage of correctly classified traces in the held-out test set.
- **AUROC**: The Area Under the Receiver Operating Characteristic curve, which measures the probe's ability to distinguish between correct and incorrect traces independent of the decision threshold. This is crucial as the class balance (correct vs. incorrect) varies significantly between datasets.

4 Experiments

4.1 Experimental Setup

We probe internal representations by extracting hidden states from five depth-proportional layers spanning early, middle, and late computation. For each layer, we compute a set of sequence-level and reasoning-focused summaries, including global averages, final-token representations, and prefix-based aggregates over the reasoning segment. These provide complementary views of how correctness information develops as the model processes its chain of thought.

Each summary is paired with a separate logistic regression classifier. This simple architecture ensures that any predictive power reflects linearly accessible information rather than probe expressiveness. We train probes on an 80/20 split of the GSM8K training set, stratified by correctness, and evaluate them on the official test set used by lm-eval-harness. We report two metrics: the ROC area, which captures overall separability, and performance at a low false-positive operating point, which reflects practical selective-verification needs.

This setup allows us to evaluate how correctness signals vary across depth, how sensitive they are to different ways of aggregating reasoning activations, and whether the strongest signals coincide across models.

4.2 Emergence of the Correctness Signal

Figure 1 summarizes probe performance across layers and feature summaries for the Qwen2.5-Math-1.5B model. All layers exhibit meaningful separability, but the strength of the signal increases with depth. Mid-depth layers already provide strong discrimination, and the final layer (layer 25) yields the most consistently high AUROC across feature types. This pattern suggests that correctness-related structure accumulates over the course of the forward pass and becomes most linearly accessible near the end of the network.

For completeness, we note that the distilled reasoning model shows the same qualitative trend, with a slightly earlier emergence of strong separability; full results are provided in Appendix A.

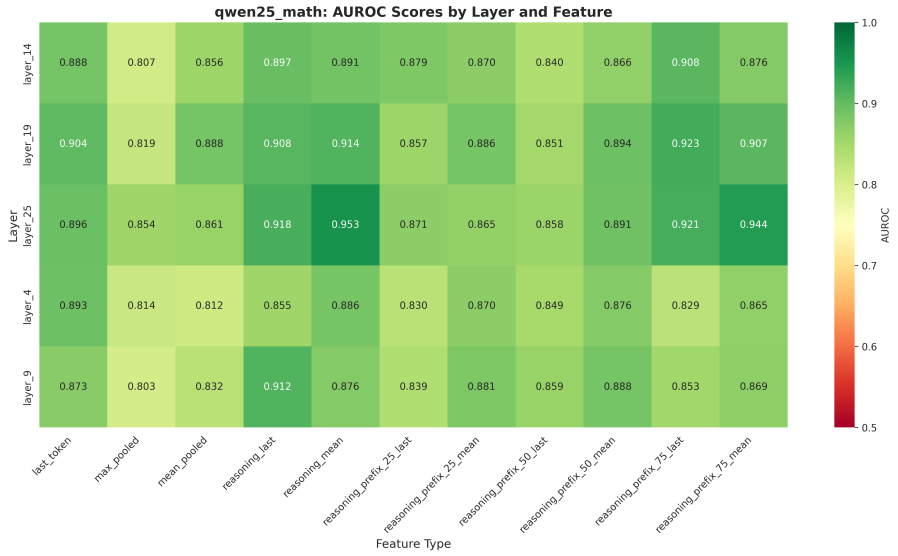


Figure 1: Probe performance across layers and feature summaries for the Qwen2.5-Math-1.5B. Darker regions indicate stronger separation between correct and incorrect trajectories.

4.3 Feature-Level Dynamics Across Depth

Figure 2 compares ROC curves for the two strongest reasoning features—reasoning-mean and reasoning-last—across layers. Both features achieve strong separability, but their behavior differs

in stability. Reasoning-mean produces smoother, more consistently steep ROC curves, especially in the low-FPR region, indicating that aggregating information across the entire reasoning span yields a more robust correctness signal. Reasoning-last also improves steadily with depth but shows greater variability, reflecting its sensitivity to local fluctuations in the final reasoning token. This advantage, however, holds primarily in-distribution; as shown in RQ3, the sequence-level last-token feature becomes more robust under distribution shift. Non-reasoning, sequence-level features such as last-token, mean-pooled, and max-pooled exhibit broadly similar in-distribution performance to mid- and late-depth reasoning features, though they tend to show less stability under distribution shift.

Across both reasoning features, deeper layers—most notably the final layer—consistently produce the cleanest and most reliable ROC curves, while mid-depth layers already perform strongly. Taken together, these results suggest that correctness information accumulates through the reasoning process and is most linearly accessible when aggregated over the full reasoning trajectory. The distilled reasoning model shows the same qualitative trend, with somewhat stronger mid-depth performance and a modestly earlier rise in separability; full results are provided in Appendix A.

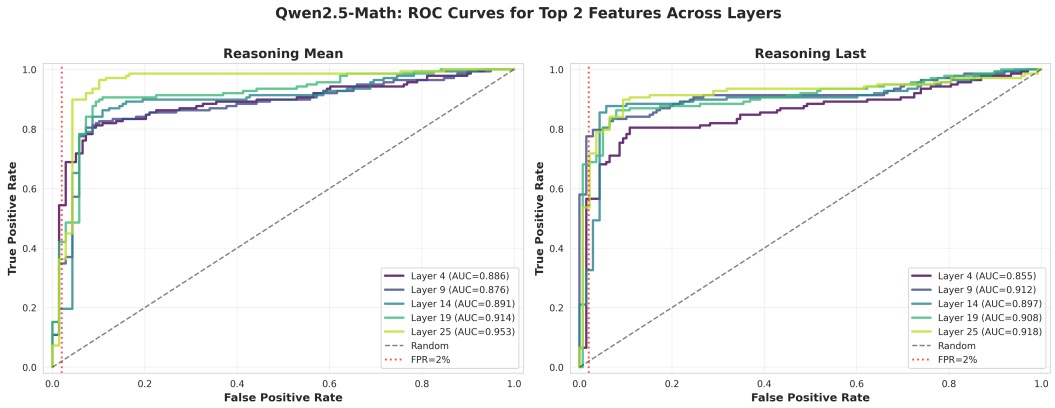


Figure 2: Representative ROC curves for the strongest features across depth in the instruction-tuned model.

4.4 Out-of-Distribution Generalization

We assess robustness by applying the probes trained on Qwen2.5-Math-1.5B - GSM8K to the algebra subset of the MATH benchmark. This dataset presents both a substantial domain shift and a dramatically different error distribution, making it a strong test of whether the internal correctness signal generalizes beyond training conditions.

Figures 3 and 4 show that, while separability drops substantially under the MATH distribution shift, several probes still retain meaningful predictive power. The transferability is uneven across layers: intermediate layers frequently outperform the deepest layer, and in some feature types the final layer degrades the most. This suggests that the upper layers of the model rely more heavily on GSM8K-specific reasoning patterns, whereas mid-depth representations capture information that generalizes more reliably across mathematical domains. This contrasts with the in-distribution setting, where the deepest layer is strongest; together these results suggest that late layers overfit GSM8K-specific reasoning patterns, while mid-depth representations encode more transferable structure.

Feature-level behavior also differs from the in-distribution setting. Across layers, the last-token feature consistently provides the strongest OOD separability, outperforming both mean-based aggregates and most reasoning-focused features. While prefix-based and max-pooled summaries exhibit reasonable stability, their ROC curves remain noticeably weaker than last-token under the MATH shift, and deeper-layer summaries often show larger degradation. These patterns indicate that surface-level features tied to the final output token transfer more reliably than broader aggregates or reasoning-span summaries, and that mid- to late-depth representations combined with last-token features provide the most robust cues under domain shift.

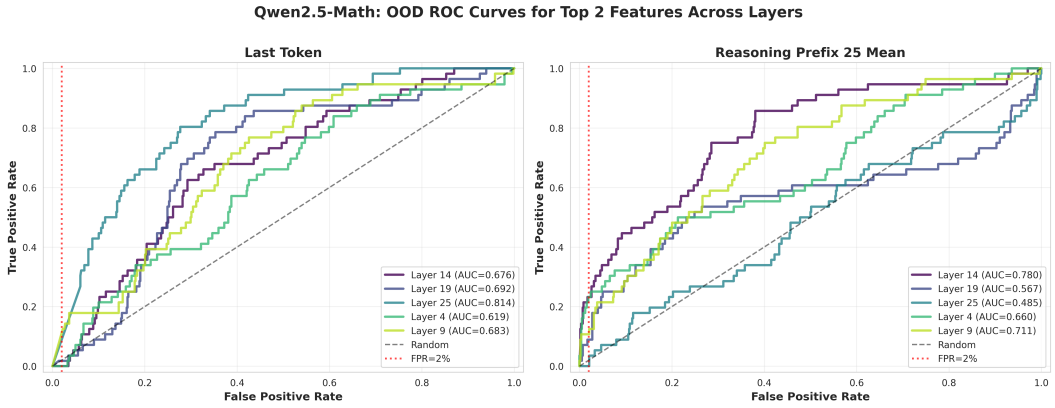


Figure 3: Out-of-distribution ROC curves for probes trained on GSM8K and evaluated on MATH. Last token has shown decent performance transferability.

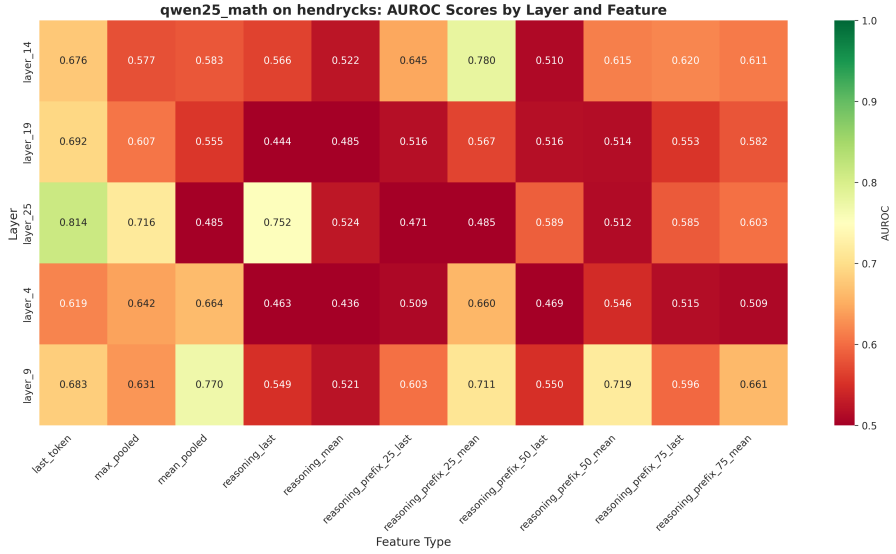


Figure 4: Out-of-distribution probe performance across layers. Only specific layers and aggregation methods performed well

4.5 Baseline Comparison

We compare our activation-based probe to a simple output-level baseline: the mean entropy of the model's token distribution during generation. Both methods use 500 GSM8K samples with an 80/20 split from 500 samples. For the entropy baseline, we compute per-token entropy from the logit distribution and average across the response to obtain one scalar per sample. For the activation probe, we extract hidden states from a layer 25) at the final reasoning token.

Figure 5 shows the ROC comparison. The entropy baseline achieves an AUROC of 0.63, while the activation probe reaches 0.77, a substantial improvement even with slightly fewer samples. The activation probe provides far better calibration and separability, supporting the value of probing internal representations for correctness prediction. This setup is small-scale but serves as a clear proof-of-concept.

4.6 R1-Distilled Qwen

As detailed in Appendix A Figure 6-7, the R1 distilled model shows the same qualitative pattern as the instruction-tuned model: correctness separability rises with depth, mid-layers already perform well,

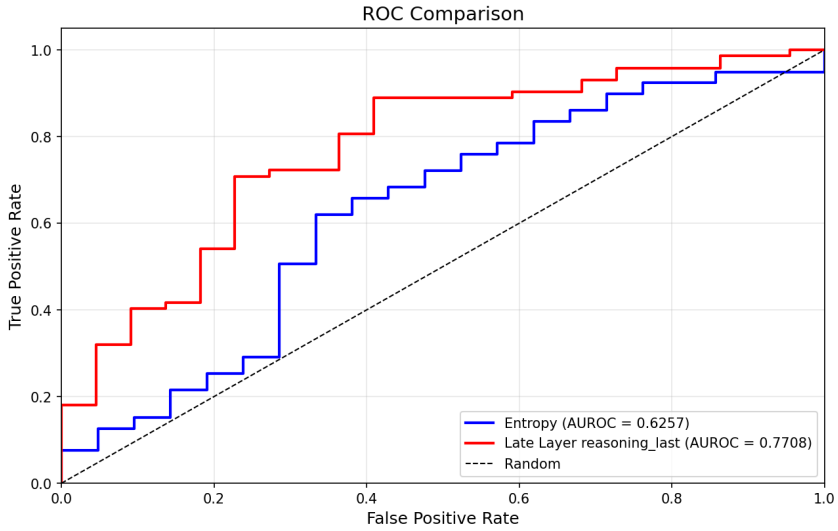


Figure 5: ROC curves for entropy baseline and late-layer activation probe.

and the final layer is strongest. The only difference is an earlier emergence of the signal, consistent with its reasoning-focused training.

5 Conclusion and Future Work

Our study shows that small reasoning models encode a clear and linearly decodable notion of correctness within their hidden representations. The signal strengthens as computation progresses, emerging earliest in the distilled reasoning model and most strongly in the final layers of the instruction-tuned model. Probes trained on GSM8K remain effective under substantial distribution shift, indicating that these correctness features generalize beyond the training domain and can support practical early-warning or selective-verification mechanisms.

By mapping where and when these signals appear, we provide a foundation for lightweight internal monitors that can assess reasoning quality without additional sampling or external verifiers. This suggests that correctness prediction is feasible even at the 1.5B scale and can serve as a building block for more reliable and efficient reasoning pipelines.

We highlight several promising directions for future work:

1. **Larger Models**: Scaling this analysis to 7B+ parameter models to see if the observed patterns hold.
2. **Harder Tasks**: Extending to code generation (HumanEval (Chen et al., 2021)) and logic puzzles where verification is strictly binary.
3. **Activation Steering**: Performance might be improved by adding the "correctness vector", using mean difference vector or attribution from classifier, to the hidden states during inference and vary the steering strength to study its effect on downstream reasoning accuracy.

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A Appendix

A.1 Implementation Details

We use the `transformers` library for model inference. All experiments are conducted on NVIDIA A100 GPUs. For probing, we use `scikit-learn`'s LogisticRegression with default parameters (L2 regularization, $C = 1.0$).

A Additional Results for Distilled Reasoning Model

The distilled reasoning model exhibits the same qualitative pattern as Qwen2.5-Math: weak discrimination in early layers, clear separation by mid-depth layers, and strongest performance in the upper portion of the network. Its correctness signal tends to emerge earlier, consistent with the model's reasoning-focused training.

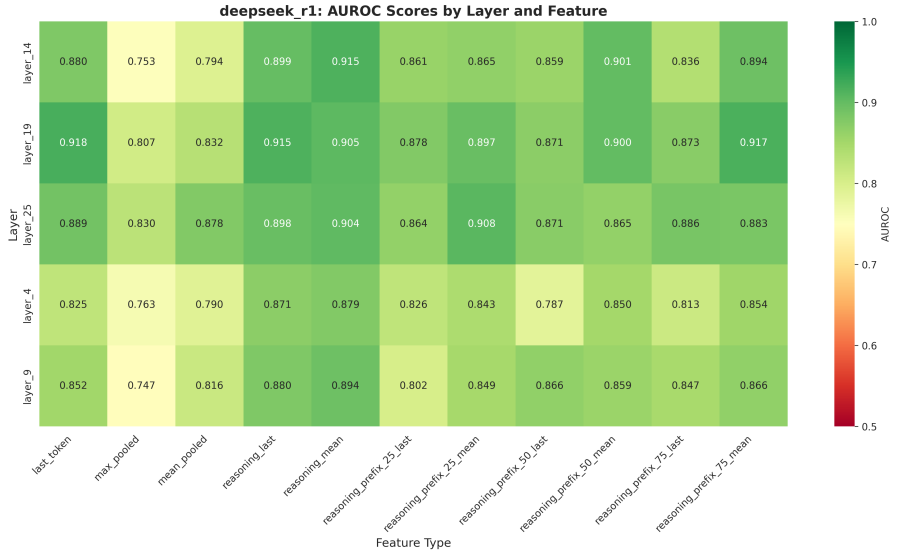


Figure 6: Probe performance across layers and feature summaries for the distilled reasoning model.

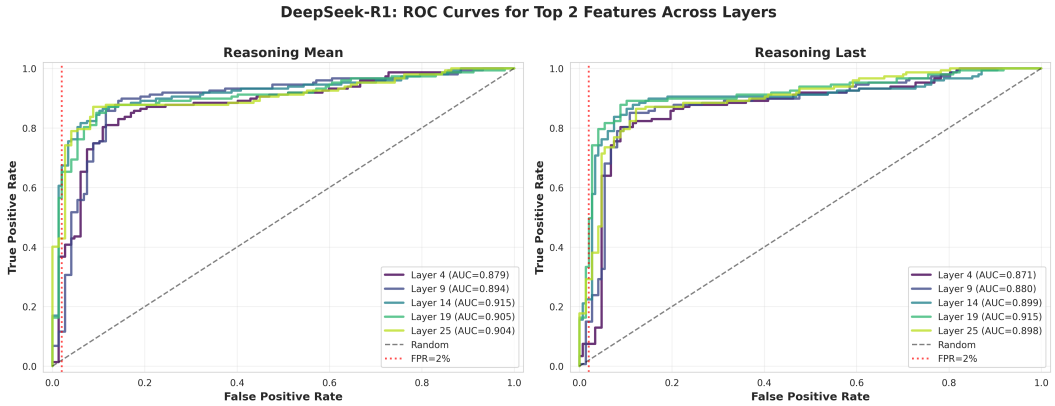


Figure 7: Representative ROC curves across depth for the distilled reasoning model.